HOSTEL MANAGEMENT MODULE (BOARDING & RESIDENCE ENGINE)

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 • March 2026
Confidential • Internal Documentation

Audience: School Administrators • Hostel Wardens • IT Team • Students • Parents
Related Modules: Student Management • Attendance • Finance • Notifications • Parent Portal • Security

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1. MODULE OVERVIEW

1.1 Purpose

The Hostel Management Module manages all aspects of student accommodation within the school. It enables:

- Dormitory and bed space management
- Student allocation and occupancy tracking
- Hostel attendance and roll call
- Leave request and approval workflows
- Daily schedule management (meals, lights-out, routines)
- Monitoring and reporting of hostel operations

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2. CORE OBJECTIVES

1. Provide centralized dormitory and bed allocation management
2. Track hostel occupancy and utilization in real time
3. Enable efficient roll call and attendance tracking
4. Manage student leave requests and approvals
5. Integrate hostel data with student and finance modules
6. Provide analytics on occupancy, attendance, and compliance
7. Ensure secure, role-based access and audit tracking
8. Support mobile-first interaction for wardens and staff

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3. REAL-WORLD SYSTEM FLOW

3.1 Lifecycle Flow

Dormitories Created → Beds Configured → Students Assigned → Occupancy Updated → Daily Attendance Recorded → Leave Requests Managed → Reports Generated

3.2 Daily Operational Flow

Warden initiates roll call
↓
Students marked present/absent/on-leave
↓
System updates attendance records
↓
Leave requests reviewed and approved/rejected
↓
Notifications sent to parents (if applicable)
↓
Daily logs and analytics updated

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4. MODULE ARCHITECTURE

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Hostel Management Module
│
├── Dormitory Management
├── Bed & Allocation Management
├── Hostel Attendance (Roll Call)
├── Leave Management
├── Daily Schedule Management
├── Analytics & Reporting
├── Notifications & Alerts
├── Parent Portal Integration
├── Mobile & Offline Support
├── Security & Role-Based Access
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5. DORMITORY MANAGEMENT

5.1 Features

- Create and manage dormitories (e.g., boys/girls/wings)
- Define dormitory capacity
- Assign wardens to dormitories

5.2 Data Model

Dormitory

- dormitory_id
- name
- type (boys/girls/mixed)
- capacity
- warden_id

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6. BED & ALLOCATION MANAGEMENT

6.1 Features

- Configure bed spaces with unique identifiers
- Assign students to specific beds
- Track occupancy in real time
- Reassign or transfer students between dormitories

6.2 Data Model

Bed

- bed_id
- dormitory_id
- bed_number
- status (available/occupied)

Allocation

- allocation_id
- student_id
- bed_id
- start_date
- status

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7. HOSTEL ATTENDANCE (ROLL CALL)

7.1 Features

- Record daily attendance (morning/evening roll call)

- Track present, absent, and on-leave students
- Mobile-based attendance marking (planned upgrade)

7.2 Rules

IF student not marked present
→ Mark as absent
→ Trigger alert (optional)

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8. LEAVE MANAGEMENT

8.1 Features

- Submit and track leave requests
- Multi-level approval workflow (warden → admin → parent)
- Record leave duration and reason
- Notify parents and administrators

8.2 Data Model

LeaveRequest

- leave_id
- student_id
- start_date
- end_date
- reason
- status (pending/approved/rejected)

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9. DAILY SCHEDULE MANAGEMENT

9.1 Features

- Define daily routines (roll call, meals, lights-out)
- Send reminders and alerts to students and wardens
- Ensure compliance with hostel rules

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10. ANALYTICS & REPORTING

10.1 Features

- Occupancy rates and trends
- Attendance summaries by dormitory
- Leave statistics and compliance tracking
- Historical reporting

10.2 Dashboards

- Hostel occupancy overview
- Attendance trends
- Leave activity reports

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11. NOTIFICATIONS & ALERTS

11.1 Features

- Alerts for absentee students
- Notifications for leave approvals/rejections
- Daily schedule reminders
- Emergency alerts

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12. PARENT PORTAL INTEGRATION

12.1 Features

- View student hostel status and attendance
- Receive notifications for leave and absences
- Communication with wardens

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13. MOBILE-FIRST & OFFLINE DESIGN

13.1 Requirements

- Mobile-friendly roll call interface
- Offline attendance capture with sync
- Push notifications for wardens and parents

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14. SECURITY & ROLE-BASED ACCESS

Role	Access Scope
Admin	Full control, reports, configuration
Warden	Manage dormitories, attendance, leave
Parent	View child's hostel data
Student	Submit leave requests, view status

14.2 Security Features

- Secure authentication and access control
- Audit logs for attendance and leave actions
- Data protection compliance

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15. SCALABILITY

- Support large boarding schools with multiple dormitories
- Multi-campus and multi-tenant support
- Efficient real-time data handling

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16. IMPLEMENTATION GAPS (CRITICAL FIXES)

16.1 Current Issues Identified

- No dormitories, beds, or student allocations configured
- Attendance system inactive due to missing data
- No analytics or reporting dashboards
- Weak integration with other modules
- No visible role-based access controls

16.2 Required Fixes

- Configure dormitories and bed spaces
- Implement student allocation workflows
- Activate attendance tracking with real data
- Build analytics dashboards
- Integrate with Student and Notification modules

- Strengthen role-based access and audit logs

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17. FUTURE ENHANCEMENTS

- Biometric or QR-based attendance
- Smart occupancy optimization
- Integration with meal management systems
- AI-based anomaly detection (e.g., missing students)
- Emergency response tracking